



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



BlackCat / ALPHV Malware Analysis - f815f5d6c85b

Malware Analysis Report

Classification	TLP:CLEAR
Published	2026-06-16
Version	1.0
Author	Lost Boys Cyber
Report Type	Malware Analysis Report

TLP:CLEAR | Lost Boys Cyber | BlackCat / ALPHV Malware Analysis - f815f5d6c85b - Malware Analysis Report

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Analysis Run Metadata

Field	Value
Timestamp	2026-06-16T18:00:25.912995+00:00
Malware Family / Case Name	BlackCat / ALPHV ransomware sample
Sample SHA256	f815f5d6c85bcb1ec071dd39532a20f5ce910989552d980d1d4346f57b75f89
Sample File	f815f5d6c85bcb1ec071dd39532a20f5ce910989552d980d1d4346f57b75f89.sample
Execution VM	win-malware-lab / snapshot clean-guestadditions-sysmon
Analysis VM	remnux
Network	vboxnet0, gateway/DNS 192.168.56.1, guest 192.168.56.20
Packet Capture	capture.pcapng

1. Executive Summary

This TRASHcan run produced an evidence-backed malware-analysis draft for f815f5d6c85bcb1ec071dd39532a20f5ce910989552d980d1d4346f57b75f89.sample using local sandbox telemetry, static triage, GhidraMCP/Ghidra-ready reverse-engineering evidence capture, bundled YARA family matching, generated detection content, deterministic scoring, and a private-client artifact bundle. Treat this output as source material for a final Lost Boys Cyber report rather than a finished client deliverable.

- Deterministic verdict: malicious
- Evidence score: 85
- Confidence: 0.85
- Scoring reasons: packer/obfuscation indicators, suspicious static strings or commands, dynamic behavioral findings, ATT&CK techniques mapped

2. Sample Metadata

Field	Value
Sample SHA256	f815f5d6c85bcb1ec071dd39532a20f5ce910989552d980d1d4346f57b75f89
Sample SHA1	45212fa4501ede5af428563f8043c4ae40faec76
Sample MD5	7a34b6a3c558492c04f3418d726b86a8
Approximate Size	3414528 bytes
File Type	/media/sf_remnux_transfer/analysis-runs/2026-06-16_175434_f815f5d6c85b/f815f5d6c85bcb1ec071dd39532a20f5ce910989552d980d1d4346f57b75f89.sample: PE32 executable (GUI) Intel 80386 (stripped to external PDB), for MS Wind...

Execution VM	win-malware-lab
Analysis Path	remnux

3. Static Analysis

This section summarizes static file characteristics, build indicators, section layout, and tool output relevant to triage and reverse-engineering follow-up.

3.1 File and Build Characteristics

- file classification: /media/sf_remnux_transfer/analysis-runs/2026-06-16_175434_f815f5d6c85b/f815f5d6c85bcb1ec071dd39532a20f5ce910989552d980d1d4346f57b75f89.sample: PE32 executable (GUI) Intel 80386 (stripped to external PDB), for MS Wind...
- Static strings output: strings.txt
- Full static triage JSON: static_triage.json

3.2 rabin2 Metadata

```

arch      x86
baddr     0x400000
binsz     3414528
bintype   pe
bits      32
canary    true
injprot   false
retguard  false
class     PE32
cmp.csum  0x00345af0
compiled  Tue Jan  4 23:04:17 2022
crypto    false
endian    little
havecode  true
hdr.csum  0x00345af0
laddr     0x0
lang      c
linenum   true
lsyms     true
machine   i386
nx        true
os        windows
overlay   false
cc        cdecl
pic       true
relocs    true
signed    false
sanitize  false
static    false
stripped  true
subsys    Windows GUI
va        true

```

3.3 Section Layout

[Sections]

nth	paddr	size	vaddr	vsize	perm	type	name
0	0x00000400	0x1f2800	0x00401000	0x1f3000	-r-x	----	.text
1	0x001f2c00	0x200	0x005f4000	0x1000	-rw-	----	.data
2	0x001f2e00	0xd7000	0x005f5000	0xd7000	-r--	----	.rdata
3	0x002c9e00	0x11e00	0x006cc000	0x12000	-r--	----	.eh_fram
4	0x00000000	0x0	0x006de000	0x1000	-rw-	----	.bss
5	0x002dbc00	0x1c00	0x006df000	0x2000	-rw-	----	.idata

6	0x002dd800	0x200	0x006e1000	0x1000	-rw- ----	.CRT
7	0x002dda00	0x64000	0x006e2000	0x95000	-rw- ----	.tls

4. Code Analysis and Embedded Artifacts

This section highlights import-level capability clues, embedded configuration artifacts, and reverse-engineering follow-up requirements.

4.1 Imports and Logic Clues

[Imports]						
nth	vaddr	bind	type	lib	name	
1	0x006df498	NONE	FUNC	ADVAPI32.dll	CloseServiceHandle	
2	0x006df49c	NONE	FUNC	ADVAPI32.dll	ControlService	
3	0x006df4a0	NONE	FUNC	ADVAPI32.dll	CreateProcessWithLogonW	
4	0x006df4a4	NONE	FUNC	ADVAPI32.dll	EnumDependentServicesW	
5	0x006df4a8	NONE	FUNC	ADVAPI32.dll	EnumServicesStatusExW	
6	0x006df4ac	NONE	FUNC	ADVAPI32.dll	GetUserNameW	
7	0x006df4b0	NONE	FUNC	ADVAPI32.dll	OpenProcessToken	
8	0x006df4b4	NONE	FUNC	ADVAPI32.dll	OpenSCManagerW	
9	0x006df4b8	NONE	FUNC	ADVAPI32.dll	OpenServiceW	
10	0x006df4bc	NONE	FUNC	ADVAPI32.dll	QueryServiceStatusEx	
11	0x006df4c0	NONE	FUNC	ADVAPI32.dll	RegCloseKey	
12	0x006df4c4	NONE	FUNC	ADVAPI32.dll	RegOpenKeyExW	
13	0x006df4c8	NONE	FUNC	ADVAPI32.dll	RegQueryValueExW	
14	0x006df4cc	NONE	FUNC	ADVAPI32.dll	RegSetValueExW	
1	0x006df4d4	NONE	FUNC	KERNEL32.dll	AcquireSRWLockExclusive	
2	0x006df4d8	NONE	FUNC	KERNEL32.dll	AcquireSRWLockShared	
3	0x006df4dc	NONE	FUNC	KERNEL32.dll	AddVectoredExceptionHandler	
4	0x006df4e0	NONE	FUNC	KERNEL32.dll	AttachConsole	
5	0x006df4e4	NONE	FUNC	KERNEL32.dll	CancelIo	
6	0x006df4e8	NONE	FUNC	KERNEL32.dll	CloseHandle	
7	0x006df4ec	NONE	FUNC	KERNEL32.dll	CompareStringOrdinal	
8	0x006df4f0	NONE	FUNC	KERNEL32.dll	CopyFileExW	
9	0x006df4f4	NONE	FUNC	KERNEL32.dll	CreateEventW	
10	0x006df4f8	NONE	FUNC	KERNEL32.dll	CreateFileMappingA	
11	0x006df4fc	NONE	FUNC	KERNEL32.dll	CreateFileW	
12	0x006df500	NONE	FUNC	KERNEL32.dll	CreateMutexA	
13	0x006df504	NONE	FUNC	KERNEL32.dll	CreateNamedPipeW	
14	0x006df508	NONE	FUNC	KERNEL32.dll	CreateProcessW	
15	0x006df50c	NONE	FUNC	KERNEL32.dll	CreateThread	
16	0x006df510	NONE	FUNC	KERNEL32.dll	CreateToolhelp32Snapshot	
17	0x006df514	NONE	FUNC	KERNEL32.dll	DeleteFileW	
18	0x006df518	NONE	FUNC	KERNEL32.dll	DeviceIoControl	
19	0x006df51c	NONE	FUNC	KERNEL32.dll	DuplicateHandle	
20	0x006df520	NONE	FUNC	KERNEL32.dll	EnterCriticalSection	
21	0x006df524	NONE	FUNC	KERNEL32.dll	ExitProcess	
22	0x006df528	NONE	FUNC	KERNEL32.dll	FileTimeToSystemTime	
23	0x006df52c	NONE	FUNC	KERNEL32.dll	FillConsoleOutputAttribute	
... (175 more lines omitted)						

4.2 Bundled YARA Triage Hits

- None observed

4.3 GhidraMCP / Ghidra Reverse-Engineering Follow-up

- Use GhidraMCP for sample-backed code-level validation whenever Ghidra is available.
- Preserve the Ghidra project path, imported program name, decompiled snippets, renamed functions/symbols, comments, xrefs, offsets, recovered config values, and any MCP transcript/summary notes under the final analyst-kit notes/ or evidence/ bundle.
- If GhidraMCP is unavailable, document the blocker and use radare2/headless Ghidra output as the fallback evidence path.

- msedge.api.cdp.microsoft.com
- nav.smartscreen.microsoft.com
- ntp.msn.com
- pqtwnoz
- pqtwnoz.local
- settings-win.data.microsoft.com
- slscr.update.microsoft.com
- v20.events.data.microsoft.com
- win-malware-lab
- win-malware-lab.local
- www.bing.com
- www.msftconnecttest.com

7.3 TLS SNI

- arc.msn.com
- checkappexec.microsoft.com
- config.edge.skype.com
- edge.microsoft.com
- fs.microsoft.com
- g.live.com
- go.microsoft.com
- msedge.api.cdp.microsoft.com
- nav.smartscreen.microsoft.com
- ntp.msn.com
- settings-win.data.microsoft.com
- slscr.update.microsoft.com
- v20.events.data.microsoft.com
- www.bing.com

7.4 HTTP Requests

- None observed

Full HTTP request count: 103

7.5 Fake HTTP/HTTPS Hits

- None observed

Full fake-service hit count: 0

7.6 Suricata Alerts

- None observed

7.7 C2 Candidate Scoring

- None observed

7.8 Memory Analysis

- Memory dump present: False
- Volatility 3 available: False
- Memory analysis JSON: memory_analysis.json

7.9 MITRE ATT&CK Mapping

- T1059.001 PowerShell: encoded/base64 PowerShell-like strings
- T1105 Ingress Tool Transfer: download or LOLBIN command strings
- T1060 persistence: Autorun registry value points to a user-writable path
- T1071 Application Layer Protocol: network communication observed

8. Full IOC Summary

- Total normalized IOC rows written to iocs_full.csv: 253
- Full machine-readable IOC coverage is stored in CSV form for downstream enrichment or upload.

8.1 High-Signal IOC and Observable Summary

Type	Observable	Source / Context
SHA256	f815f5d6c85bcb1ec071dd39532a20f5ce910989552d980d1d4346f57b75f89	Primary submitted sample
SHA1	45212fa4501ede5af428563f8043c4ae40faec76	Static triage
MD5	7a34b6a3c558492c04f3418d726b86a8	Static triage
Onion service	2cuqgeerjdba2rhdiviezodpu3lc4qz2sjf4qin6f7std2evleqlzjid.onion	Embedded ransom portal / recovery workflow
Ransom note URL pattern	http://2cuqgeerjdba2rhdiviezodpu3lc4qz2sjf4qin6f7std2evleqlzjid.onion/?access-key=\${ACCESS_KEY}	Embedded configuration string
Ransom note filename pattern	RECOVER-#{EXTENSION}-FILES.txt	Embedded configuration string
Encrypted-file extension	wpzlbji	Embedded configuration string
Host utility reference	vssadmin.exe	Static strings; anti-recovery / shadow-copy tradecraft indicator
Registry persistence candidate	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\OneDrive	Sandbox persistence observation; validate before actioning as malicious

8.2 IOC Handling Notes

The full normalized IOC export remains available in iocs_full.csv with 253 rows. The table above intentionally prioritizes high-signal, case-relevant observables for the client report and avoids flooding the PDF with low-confidence parser artifacts such as Rust library paths, browser cache files, and benign Microsoft connectivity domains.

9. Detection Engineering

This section preserves detection artifacts generated from the sandbox run and provides analyst-ready content for endpoint and file-based detection workflows.

TRASHcan generated the following detection artefacts automatically:

- rule.yar
- yara_triage_summary.json
- yara_triage_hits.txt
- process_tree_summary.md
- iocs_full.csv
- sigma_behavior.yml

9.1 Detection Content

`sigma_behavior.yml`

```
title: Sandbox Observed Autorun Persistence To User Writable Path f815f5d6c85b
id: 30b0b72a-089b-558b-9e03-d43fa030ab48
```

```
{
    rule RAIconoon_f815f5d6c85b_Triage
{
    meta:
        description = "Auto-generated triage rule for local sandbox run"
        sha256 = "f815f5d6c85bcbec071dd39532a20f5ce910989552d980d1d4346f57b75f89"
        author = "RAIconoon local sandbox"

    strings:
        $s1 = {"\config_id\":"\", \"public_key\":
\"MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAugZ4ATE9+9FqununW/DBvGosnUX/
bNxQzMYUmE14GIbNa6vwYSNXOLG09mvdAqZqD3LXihWDJy25+gzqSeS+Fs2qNyTdFGPA8iu2xx5RRUXKLGFThxtIzg3fohAK3+LxJV
hxtuITAT38IHacc7dVLHsrddu4UDjiHGFDvXjB55Nwe5cu1BYylHsARMYycBA2FWlP57cKvc2/C30XBAF6qbsVXBcyFhrKO0YA/
+5IJfFEhgHy2FLHRf8lmPQPbSlrM6dk+W4D5KVQPx/eFp0geUJJmlre3flI29qWS20bkGqAEz9j07y69HGYN9nt7+DRGbwrpNo/
EkZkuaSttQIDAQAB\", \"extension\":\"wpzlbji\", \"note_file_name\":\"RECOVER-{$[EXTENSION]-FILES.txt\",
\"note_full_text\":\">> Introduction\\n\\nImportant files on your system was ENCRYPTED and now they
have have \\n\\n{$[EXTENSION]}\\n\\n extension.\\n\\nIn order to recover your files you need to follow
instructions below.\\n\\n>> Sensitive Data\\n\\nSensitive data on your system was DOWNLOADED and it
will be PUBLISHED if you refuse to cooperate.\\n\\nData includes:\\n- Employees personal data, CVs, DL,
SSN.\\n- Complete network map including credentials for local and remote services.\\n- Financial
information including clients data, bills, budgets, annual reports, bank statements.\\n- Complete
datagrams/schemas/drawings for manufacturing in solidworks format\\n- And more...\\n\\n>> CAUTION\\n\\
\\nDO NOT MODIFY FILES YOURSELF.\\nDO NOT USE THIRD PARTY SOFTWARE TO RESTORE YOUR DATA.\\nYOU MAY
DAMAGE YOUR FILES, IT WILL RESULT IN PERMANENT DATA LOSS.\\nYOUR DATA IS STRONGLY ENCRYPTED, YOU CAN
NOT DECRYPT IT WITHOUT CIPHER KEY.\\n\\n>> Recovery procedure\\n\\nFollow these simple steps to get in
touch and recover your data:\\n1) Download and install Tor Browser from: https://torproject.org/\\n2)
Navigate to: http://2cuqgeerjdba2rhdiviezodpu3lc4qz2sfj4qin6f7std2evleqlzjid.onion/?access-key=$
{ACCESS_KEY}\"}, \"note_short_text\":\"Important files on your system was ENCRYPTED.\\nSensitive data on
your system was DOWNLOADED.\\nTo recover your files and prevent publishing of sensitive information
follow instructions in \\n\\n{$NOTE_FILE_NAME}\\n\\n file.\", \"default_file_mode\":{\"SmartPattern\":
[10485760,10]}, \"default_file_cipher\":\"Best\", \"credentials\":[[\"\", \"\"], [\"\", \"\"], [\"\", \"\"],
[\"\", \"\"]], \"kill_services\":[\"mepocs\", \"memtas\", \"veeam\", \"svc$\", \"backup\", \"sql\", \"vss\",
```

```

\\msexchange\\",\\kill_processes\\":[\\encsvc\\",\\thebat\\",\\mydesktopqos\\",\\xfssvccon\\",\\firefox\\",
\\infopath\\",\\winword\\",\\steam\\",\\synctime\\",\\notepad\\",\\ocomm\\",\\onenote\\",\\mspub\\",
\\thunderbird\\",\\agntsvc\\",\\sql\\",\\excel\\",\\powerpnt\\",\\outlook\\",\\wordpad\\",\\dbeng50\\",
\\isqlplussvc\\",\\sqbcoreservice\\",\\oracle\\",\\ocautoupds\\",\\dbsnmp\\",\\msaccess\\",\\tbirdconfig\\",
\\ocssd\\",\\mydesktopservice\\",\\visio\\"],\\exclude_directory_names\\":[\\system volume information\\",
\\intel\\",\\$windows.~ws\\",\\application data\\",\\$recycle.bin\\",\\mozilla\\",\\program files (x86)\\",
\\program files\\",\\$windows.b\\",\\public\\",\\msocache\\",\\windows\\",\\default\\",\\all users\\",\\tor
browser\\",\\programdata\\",\\boot\\",\\config.msi\\",\\google\\",\\perflogs\\",\\appdata\\",\\windows.old\\"],
\\exclude_file_names\\":[\\desktop.ini\\",\\autorun.inf\\",\\ntldr\\",\\bootsect.bak\\",\\thumbs.db\\",
\\boot.ini\\",\\ntuser.dat\\",\\iconcache.db\\",\\bootfont.bin\\",\\ntuser.ini\\",\\ntuser.dat.log\\"],
\\exclude_file_extensions\\":[\\themepack\\",\\nls\\",\\diagpkg\\",\\msi\\",\\lnk\\",\\exe\\",\\cab\\",\\scr\\",
\\bat\\",\\drv\\",\\rtp\\",\\msp\\",\\prf\\",\\msc\\",\\ico\\",\\key\\",\\ocx\\",\\diagcab\\",\\diagcfg\\",
\\pdb\\",\\wp\\",\\hlp\\",\\icns\\",\\rom\\",\\dll\\",\\msstyles\\",\\mod\\",\\ps1\\",\\ics\\",\\hta\\",\\bin\\",
\\cmd\\",\\ani\\",\\386\\",\\lock\\",\\cur\\",\\idx\\",\\sys\\",\\com\\",\\deskthemepack\\",\\shs\\",\\ldf\\",
\\theme\\",\\mpa\\",\\nomedia\\",\\spl\\",\\cpl\\",\\adv\\",\\icl\\",\\msu\\"],\\exclude_file_path_wildcard\\":
[],\\enable_network_discovery\\":true,\\enable_self_propagation\\":true,\\enable_set_wallpaper\\":true,
\\enable_esxi_vm_kill\\":true,\\enable_esxi_vm_snapshot_kill\\":false,\\strict_include_paths\\":[]}
src/bin/encrypt_app/app.rs" ascii wide
    $s2 = "CreateProcessWithLogonW=success," ascii wide
    $s3 = "CreateProcessWithLogonW=failed," ascii wide
    $s4 = "CreateProcessWithLogonW" ascii wide
    $s5 = "CreateProcessW" ascii wide
condition:
    uint16(0) == 0x5a4d and 2 of them
}

```

10. Threat Hunting

- Hunt for the bundled YARA family themes in endpoint telemetry using `qkl_triage_hunts.kql`.
- Pivot from suspicious domains, direct-IP connections, dropped files, autoruns, and process-tree anomalies captured in this run.
- If the sample behaved like a loader or access implant, scope adjacent hosts for similar parent-child chains and follow-on persistence.

10.1 Microsoft Defender XDR Hunt Queries

```

// BlackCat / ALPHV sample hash sighting
DeviceFileEvents
| where SHA256 =~ "f815f5d6c85bc91ec071dd39532a20f5ce910989552d980d1d4346f57b75f89"
   or SHA1 =~ "45212fa4501ede5af428563f8043c4ae40faec76"
   or MD5 =~ "7a34b6a3c558492c04f3418d726b86a8"
| project Timestamp, DeviceName, FolderPath, FileName, SHA256, InitiatingProcessFileName,
InitiatingProcessCommandLine

// Ransom note / onion recovery portal indicators
DeviceFileEvents
| where FileName has "RECOVER-" and FileName ends with "-FILES.txt"
   or AdditionalFields has "2cuqgeerjdba2rhdiviezodpu3lc4qz2sjf4qin6f7std2evleqlzjid.onion"
| project Timestamp, DeviceName, FolderPath, FileName, InitiatingProcessFileName,
InitiatingProcessCommandLine

// Autorun persistence pointing into user-writable locations
DeviceRegistryEvents
| where RegistryKey has_any ("\\Run", "\\RunOnce", "\\StartupApproved")
| where RegistryValueData has_any ("\\AppData\\", "\\Temp\\", "\\ProgramData\\", "\\Startup\\")
| project Timestamp, DeviceName, RegistryKey, RegistryValueName, RegistryValueData,
InitiatingProcessFileName, InitiatingProcessCommandLine

```

11. Guest Artifact Inventory

Category	Representative Artifacts	Analyst Use
Event logs	application.evtx, powershell_operational.evtx	Timeline validation and script/process telemetry review
Registry exports	hkcu_run.reg, hkln_run.reg,	Persistence validation and autorun triage

	hklm_runonce.reg	
Browser / user activity	Edge history, cache, crashpad, session-storage, and recent-link artifacts	Separate malware-created evidence from normal Windows/Edge background noise
OneDrive / Windows background telemetry	OneDrive logs, TokenBroker cache, Windows notification/search artifacts	Baseline noise suppression; do not treat as malicious without corroboration
Temp / dropped files	C__Users_analyst_AppData_Local_Temp_*.tmp, .ses, and staged dropped-file captures	Candidate payload staging and follow-up hash triage

The complete guest artifact set is preserved in guest_artifacts.zip and the run directory. The PDF intentionally summarizes the inventory so the final report remains readable while keeping the raw evidence available for validation.

12. Guest Telemetry Setup

- guest_setup.ps1 prepares Sysmon/Defender settings inside the clean Windows snapshot.
- guest_collect.ps1 exports EVTX/process/network/registry artifacts once Guest Control is available or when run manually in the guest.
- behavior_summary.json contains parsed guest-side persistence, dropped-file, and Sysmon summaries when artifacts are available.

13. Private Client Bundle and Guardrails

- Canonical report bundle root: reports/, static/, dynamic/, network/, memory/, detections/, and evidence/ under this run directory.
- Public sample/artifact enrichment is disabled by default in run_manifest.json.
- Default handling is private single-client / TLP:AMBER; do not upload live samples or artifacts to public services without explicit authorization.

14. Recommendations

- Keep source artifacts and intermediate analysis output in the run/analyst-kit area; copy only final client-ready reports into RAIconn_Reports.
- Treat public enrichment, public uploads, and third-party sharing as opt-in only after explicit client authorization.
- Use report_preflight.json, workflow_status.json, and chain-of-custody artifacts as QA evidence before delivery.

15. Analyst Notes and Next Steps

- DNS is wildcarded to 192.168.56.254 by dnsmasq.
- HTTP ports 80/8080 and HTTPS port 443 are simulated locally when available.
- Guest Control is used when available; mounted ISO and keyboard injection remain as fallback.
- This formatted report is derived from the local TRASHcan sandbox run and should be reviewed with the preserved evidence bundle before operational use.